SYSC 5108 Exam Study
Assignment 4, Question 1

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Question Summary

The network is a simple recurrent neural network with:

- two input neurons,
- three hidden ReLU neurons,
- two output ReLU neurons,
- all bias terms equal to 0.1.

Given

$$x_t = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, \qquad h_{t-1} = \begin{bmatrix} 0.05 \\ 0.2 \\ 0.15 \end{bmatrix},$$

the question asks for:

(a) the output y_t,

(b) the δ value for each neuron when the target is $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Course and Textbook Cross-References

- **Chapter 4 slides, Forward Propagation Generalization:** each layer computes an affine transformation followed by an activation.
- **Chapter 4 slides, Hidden Units:** ReLU is

$$g(z) = \max(0, z).$$

- **Chapter 4 slides, Chain Rule and Backward Propagation:** gradients propagate backward through the network using the chain rule.
- **Goodfellow, Bengio, and Courville, *Deep Learning*:** feedforward layers use affine transformations plus activations, while recurrent networks extend this by feeding a previous hidden state into the current hidden update. ReLU units remain elementwise $g(z) = \max(0, z)$.

Given Parameters

Bias vectors:

$$b_h = \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix}, \qquad b_y = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}.$$

Weight matrices:

$$W_{xh} = \begin{bmatrix} -0.07 & 0.05 \\ -0.04 & 0.10 \\ -0.05 & -0.05 \end{bmatrix},$$

$$W_{hh} = \begin{bmatrix} -0.22 & -0.10 & 0.05 \\ -0.04 & -0.09 & -0.06 \\ -0.09 & 0 & -0.08 \end{bmatrix},$$

$$W_{hy} = \begin{bmatrix} 0.06 & -0.01 & -0.18 \\ -0.06 & 0.13 & 0.14 \end{bmatrix}.$$

Step-by-Step Solution

Part (a): Compute y_t

For one recurrent step, the hidden pre-activation is

$$z_h = W_{xh}x_t + W_{hh}h_{t-1} + b_h.$$

1. Compute $W_{xh}x_t$

$$\begin{aligned} W_{xh}x_t &= \begin{bmatrix} -0.07 & 0.05 \\ -0.04 & 0.10 \\ -0.05 & -0.05 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \\ &= \begin{bmatrix} -0.07(1) + 0.05(0.5) \\ -0.04(1) + 0.10(0.5) \\ -0.05(1) - 0.05(0.5) \end{bmatrix} \\ &= \begin{bmatrix} -0.045 \\ 0.010 \\ -0.075 \end{bmatrix}. \end{aligned}$$

2. Compute $W_{hh}h_{t-1}$

$$\begin{aligned} W_{hh}h_{t-1} &= \begin{bmatrix} -0.22 & -0.10 & 0.05 \\ -0.04 & -0.09 & -0.06 \\ -0.09 & 0 & -0.08 \end{bmatrix} \begin{bmatrix} 0.05 \\ 0.2 \\ 0.15 \end{bmatrix} \\ &= \begin{bmatrix} -0.22(0.05) - 0.10(0.2) + 0.05(0.15) \\ -0.04(0.05) - 0.09(0.2) - 0.06(0.15) \\ -0.09(0.05) + 0(0.2) - 0.08(0.15) \end{bmatrix} \\ &= \begin{bmatrix} -0.0235 \\ -0.0290 \\ -0.0165 \end{bmatrix}. \end{aligned}$$

3. Add the bias and apply ReLU

$$\begin{aligned} z_h &= \begin{bmatrix} -0.045 \\ 0.010 \\ -0.075 \end{bmatrix} + \begin{bmatrix} -0.0235 \\ -0.0290 \\ -0.0165 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} \\ &= \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix}. \end{aligned}$$

Since all entries are positive,

$$h_t = \mathrm{ReLU}(z_h) = \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix}.$$

4. Compute the output layer

The output pre-activation is

$$z_y = W_{hy} h_t + b_y.$$

$$\begin{aligned} z_y &= \begin{bmatrix} 0.06 & -0.01 & -0.18 \\ -0.06 & 0.13 & 0.14 \end{bmatrix} \begin{bmatrix} 0.0315 \\ 0.0810 \\ 0.0085 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} \\ &= \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}. \end{aligned}$$

Again, both entries are positive, so ReLU leaves them unchanged:

$$y_t = \mathrm{ReLU}(z_y) = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}.$$

$$\boxed{y_t = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}}$$

Part (b): Compute the δ values

Let the target be

$$t = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Use the same convention as earlier assignments:

$$\delta_j = \frac{\partial E}{\partial z_j}.$$

Since every hidden and output pre-activation is positive, all ReLU derivatives are

$$\mathrm{ReLU}'(z) = 1.$$

1. Output-layer deltas

For each output neuron,

$$\delta = (y - t) \odot \mathrm{ReLU}'(z_y).$$

Therefore,

$$\delta_y = \begin{bmatrix} 0.09955 - 0 \\ 0.10983 - 1 \end{bmatrix} = \begin{bmatrix} 0.09955 \\ -0.89017 \end{bmatrix}.$$

So the two output-neuron deltas are

$$\boxed{\delta_6 = 0.09955, \qquad \delta_7 = -0.89017}.$$

2. Hidden-layer deltas

The hidden deltas at time t are

$$\delta_h = \left(W_{hy}^T \delta_y\right) \odot \mathrm{ReLU}'(z_h).$$

First compute $W_{hy}^T \delta_y$:

$$\begin{aligned} W_{hy}^T \delta_y &= \begin{bmatrix} 0.06 & -0.06 \\ -0.01 & 0.13 \\ -0.18 & 0.14 \end{bmatrix} \begin{bmatrix} 0.09955 \\ -0.89017 \end{bmatrix} \\ &= \begin{bmatrix} 0.0593832 \\ -0.1167176 \\ -0.1425428 \end{bmatrix}. \end{aligned}$$

Because every hidden ReLU derivative is 1,

$$\delta_h = \begin{bmatrix} 0.0593832 \\ -0.1167176 \\ -0.1425428 \end{bmatrix}.$$

Thus the hidden-neuron deltas are

$$\boxed{\delta_3 = 0.05938, \qquad \delta_4 = -0.11672, \qquad \delta_5 = -0.14254}.$$

Final Answer

$$\boxed{y_t = \begin{bmatrix} 0.09955 \\ 0.10983 \end{bmatrix}}$$

$$\boxed{\delta_3 = 0.05938, \quad \delta_4 = -0.11672, \quad \delta_5 = -0.14254, \quad \delta_6 = 0.09955, \quad \delta_7 = -0.89017}$$

Exam Notes

- This is a *single-time-step* recurrent calculation: h_{t-1} is given, so part (a) is just the hidden-state update followed by the output-layer computation.
- The hidden update has the standard RNN form

$$h_t = \phi(W_{xh} x_t + W_{hh} h_{t-1} + b_h).$$

- Since every pre-activation is positive, every ReLU derivative is 1, which makes the δ calculations especially simple.
- For this question, the hidden deltas are computed from the output layer at the same time step. No future-time contribution is needed because only one target/output step is being evaluated here.